OSI LAYER

Course: Diploma in Cyber Security

Topic: OSI 7-Layer Model

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Introduction

The OSI model is used to understand how data moves from one device to another over a network. It has 7 layers, and each layer has a different job. In this assignment, I have covered the OSI layers, common networking examples, and some simple real-life situations.

Task 1: OSI 7-Layer Model Diagram

The OSI model has seven layers. Draw the following layers from Layer 7 at the top to Layer 1 at the bottom. Add the example beside each layer.

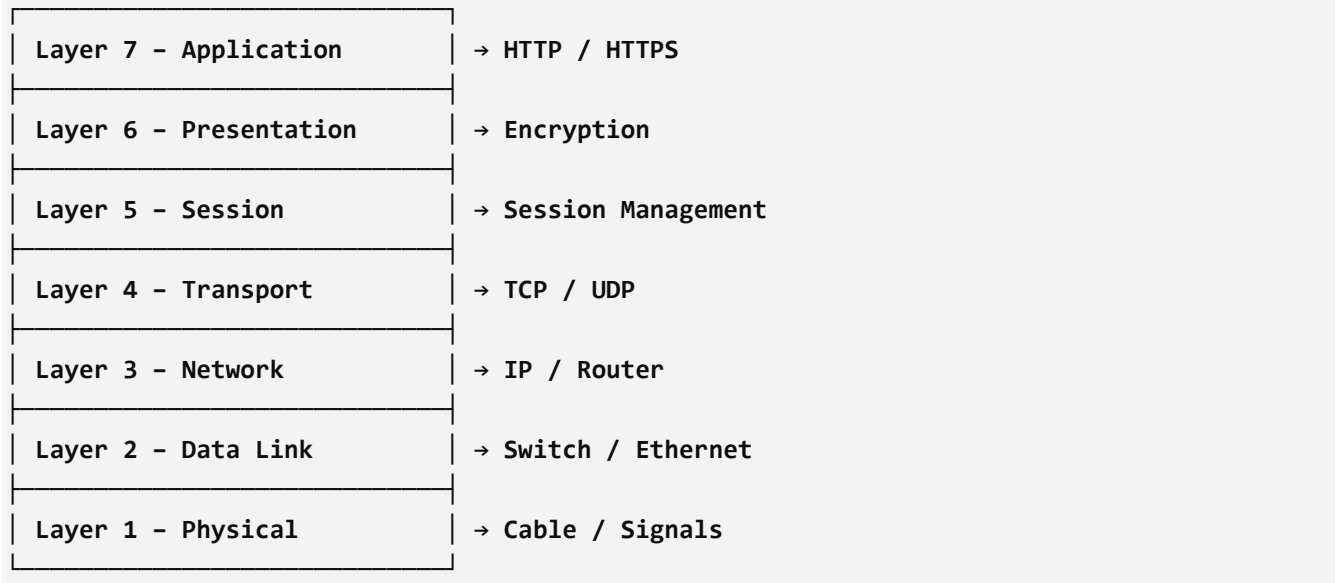


Figure Caption: Figure 1: OSI 7-Layer Model

Task 2: Match the Networking Scenarios

a) Encrypting data before sending : Presentation Layer (Layer 6)

- This layer handles encryption, decryption and data formatting.

b) Routing packets across networks : Network Layer (Layer 3)

- This layer uses IP addressing and routing to move packets between networks.

c) Converting data to electrical signals : Physical Layer (Layer 1)

- This layer sends bits as electrical, light or radio signals.

d) Opening a web page in your browser : Application Layer (Layer 7)

- This is the layer where network applications such as HTTP/HTTPS work.

e) Detecting and correcting errors in frames : Data Link Layer (Layer 2)

- This layer checks frames and can detect transmission errors on the local link.

Task 3: OSI Layers in a Daily App

App used: WhatsApp

When I send a WhatsApp message, different OSI layers work together in the background. The Application Layer handles the app data and messaging service. The Transport Layer helps

send the data using TCP or other transport methods, while the Network Layer uses IP to move packets towards the destination. The Data Link and Physical layers then carry the data through Wi-Fi, mobile network, cables or radio signals.

Phone → Wi-Fi/Mobile Network → Internet → WhatsApp Server

Task 4: Website Not Opening but Ping Works

Scenario: I can successfully ping the website IP address, but the website does not open in my browser.

If ping works, the basic network connection and IP communication are working, so the problem is less likely to be at the Network Layer. The problem can be at higher layers, especially the Transport or Application layer. For example, TCP/port 80 or 443 may be blocked, the web server may not be responding, or there may be an HTTP/HTTPS problem. DNS can also be involved when using a domain name instead of the IP address.

Simple conclusion: Ping only proves that ICMP communication to the IP address is working. It does not prove that the web service itself is working.

Conclusion

In this assignment, I learned the purpose of all 7 OSI layers and how different network tasks fit into them. I also understood that different layers handle different jobs such as

routing, transport, encryption, applications and physical signals. This model makes it easier to understand and troubleshoot network problems.

Quick Revision

- Layer 7 - Application: HTTP/HTTPS and user network applications.
- Layer 6 - Presentation: Encryption and data formatting.
- Layer 5 - Session: Starts and manages communication sessions.
- Layer 4 - Transport: TCP/UDP and end-to-end delivery.
- Layer 3 - Network: IP addressing and routing.
- Layer 2 - Data Link: Frames, MAC addressing and local error checking.
- Layer 1 - Physical: Cables, radio and electrical/light signals.